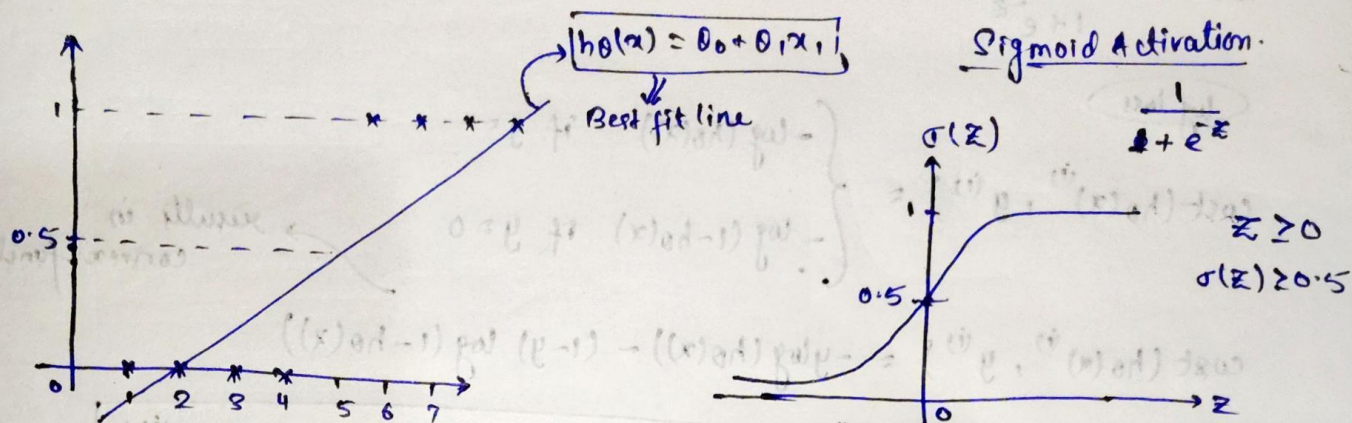


* How Logistic Regression Solves Classification problems



$$h_0(x) = \sigma(\theta_0 + \theta_1 x_1) \Rightarrow z = \theta_0 + \theta_1 x_1$$

$$= \sigma(z) \Rightarrow h_0(x) = \frac{1}{1 + e^{-z}} \sim \text{Logistic Regression hypothesis}$$

Linear Regression cost function:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2 \Rightarrow \text{convex function}$$

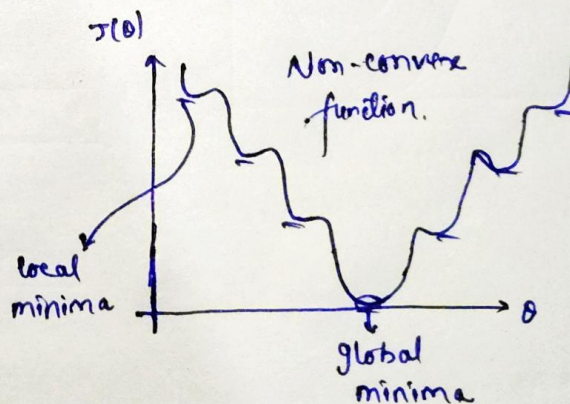
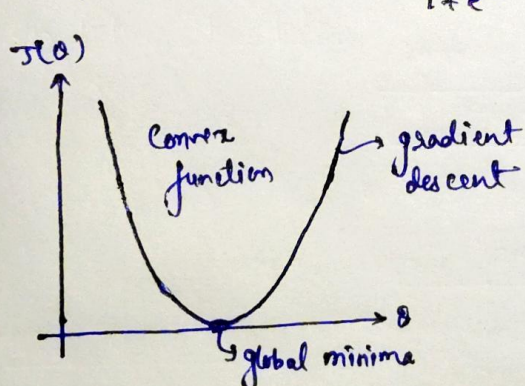
$$h_0(x) = \theta_0 + \theta_1 x$$

Logistic Regression Cost function:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2 \Rightarrow \text{Non convex functions}$$

Sigmoid Activation

$$h_0(x) = \frac{1}{1 + e^{-z}} \quad z = \theta_0 + \theta_1 x_1$$



$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2$$

Let's denote $\Rightarrow \text{cost}(h_0(x^{(i)}), y^{(i)})$

$$h_0(x)^{(1)} = \frac{1}{1 + e^{-z}}$$

$$z = \theta_0 + \theta_1 x$$

log loss

$$\text{cost}(h_0(x)^{(1)}, y^{(1)}) = \begin{cases} -\log(h_0(x)) & \text{if } y = 1 \\ -\log(1 - h_0(x)) & \text{if } y = 0 \end{cases}$$

results in convex function.

$$\text{cost}(h_0(x)^{(1)}, y^{(1)}) = -y \log(h_0(x)) - (1-y) \log(1 - h_0(x))$$

$$J(\theta_0, \theta_1) = -\frac{1}{2m} \sum_{i=1}^m (y^{(i)} \log(h_0(x)^{(1)}) - (1-y^{(i)}) \log(1 - h_0(x)^{(1)}))$$

Minimize cost function $J(\theta_0, \theta_1)$ by changing θ_0 & θ_1

Convergence Algorithm.

Repeat

$$\left\{ \begin{array}{l} \theta_j \leftarrow \theta_j - \alpha \frac{\partial J(\theta_0, \theta_1)}{\partial \theta_j} \end{array} \right. \quad j = 0 \text{ and } 1.$$

}